



Submitted By : Maryam Rehan

Cyber Security Internship Report

Week 01 Cybersecurity Testing Lab Environment Setup

Objective:

The objective of this task was to set up a cybersecurity testing lab environment using Oracle VirtualBox and Kali Linux. The lab was configured for ethical hacking and cybersecurity practice with a custom NAT Network using the 10.0.0.0/24 subnet.

Lab Setup Description:

Oracle VirtualBox was used as the base virtualization platform, and Kali Linux 2026.2 was configured as the attacking and testing machine.

A custom NAT Network named Nat Network was created using the 10.0.0.0/24 subnet. The Kali Linux virtual machine was connected to this NAT Network.

Kali Linux IP Configuration:

Kali Linux was configured with the required manual IP address:

10.0.0.2/24

The network interface was checked using the ip a command, and the required IP address was successfully assigned to the eth0 interface.

Internet Connectivity:

Internet connectivity was verified using the following command:

ping -c 4 8.8.8.8

The test was successful with 4 packets transmitted, 4 packets received, and 0% packet loss. This confirmed that Kali Linux had full Internet access.

Virtual Machine Settings:

The VirtualBox settings for Kali Linux were configured according to the lab requirements.

The Shared Clipboard was set to Bidirectional, and Drag and Drop was also set to Bidirectional. These settings allow communication and file transfer between the host machine and the virtual machine.

Shared Folder:

A shared folder named downloads was configured in the Kali Linux virtual machine. The host machine's Downloads folder was selected as the shared folder, and Auto-mount was enabled.

Snapshot:

After completing the required configuration, a snapshot of the Kali Linux virtual machine was created to preserve the completed working state.

The snapshot was named:

Kali-Lab-Setup-Complete

Final Status:

The required Phase 1 cybersecurity lab setup was successfully completed. The NAT Network, Kali Linux IP address, Internet connectivity, Clipboard, Drag and Drop, Shared Folder, and VM Snapshot were all configured according to the provided requirements.

Screenshots:

```
root@kali: ~
Session Actions Edit View Help

(root@kali)-[~]
# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86296sec preferred_lft 86296sec
    inet6 fd17:625c:f037:2:b97a:5bfb:4dfa:e0db/64 scope global dynamic noprefixroute
        valid_lft 86297sec preferred_lft 14297sec
    inet6 fe80::fc10:7c36:5c40:5dc2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether f2:96:28:c4:f7:f4 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
```

```
(root@kali)-[~]
# ls -l /etc/resolv.conf
-rw-r--r-- 1 root root 110 Sep 25 15:56 /etc/resolv.conf

(root@kali)-[~]
# cat /etc/resolv.conf
# Generated by NetworkManager
search www.mtlinkwifi.com
nameserver 192.168.0.1
nameserver fd17:625c:f037:2::3
```

```
(root@kali)-[~]
# printf "nameserver 8.8.8.8\nnameserver 1.1.1.1\n" > /etc/resolv.conf

(root@kali)-[~]
# cat /etc/resolv.conf
nameserver 8.8.8.8
nameserver 1.1.1.1

(root@kali)-[~]
# ping -c 4 google.com
PING google.com (142.250.203.14) 56(84) bytes of data:
64 bytes from pnfjra-ak-in-f14.1e100.net (142.250.203.14): icmp_seq=1 ttl=64 time=21.8 ms
64 bytes from pnfjra-ak-in-f14.1e100.net (142.250.203.14): icmp_seq=2 ttl=64 time=20.7 ms
64 bytes from pnfjra-ak-in-f14.1e100.net (142.250.203.14): icmp_seq=3 ttl=64 time=37.6 ms
64 bytes from pnfjra-ak-in-f14.1e100.net (142.250.203.14): icmp_seq=4 ttl=64 time=19.2 ms

— google.com ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 19.215/24.835/37.568/7.409 ms

(root@kali)-[~]
#
```

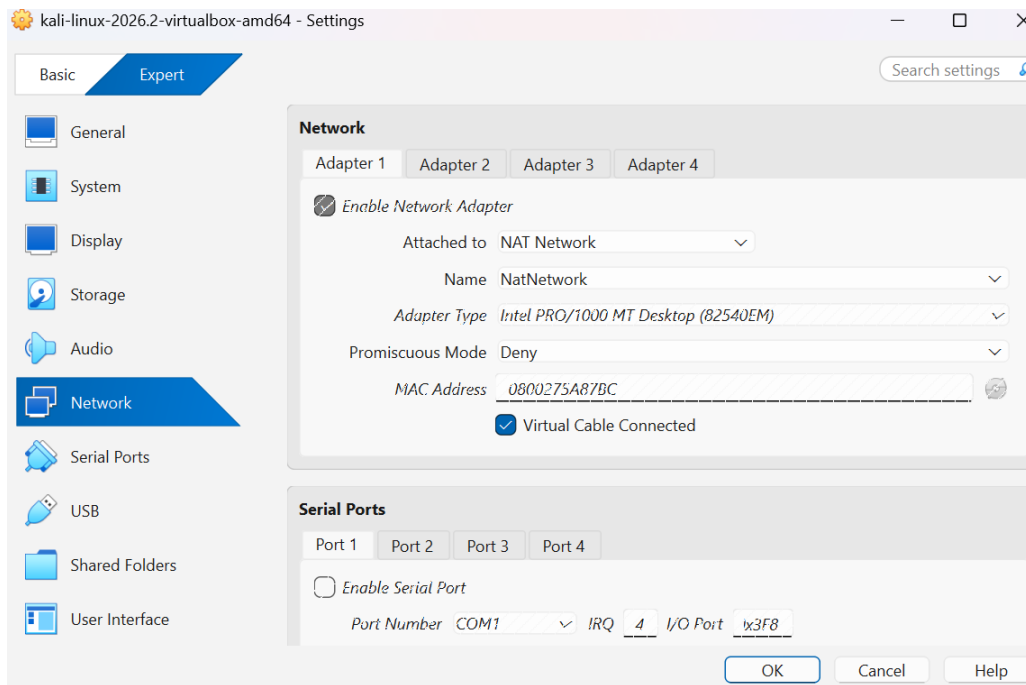
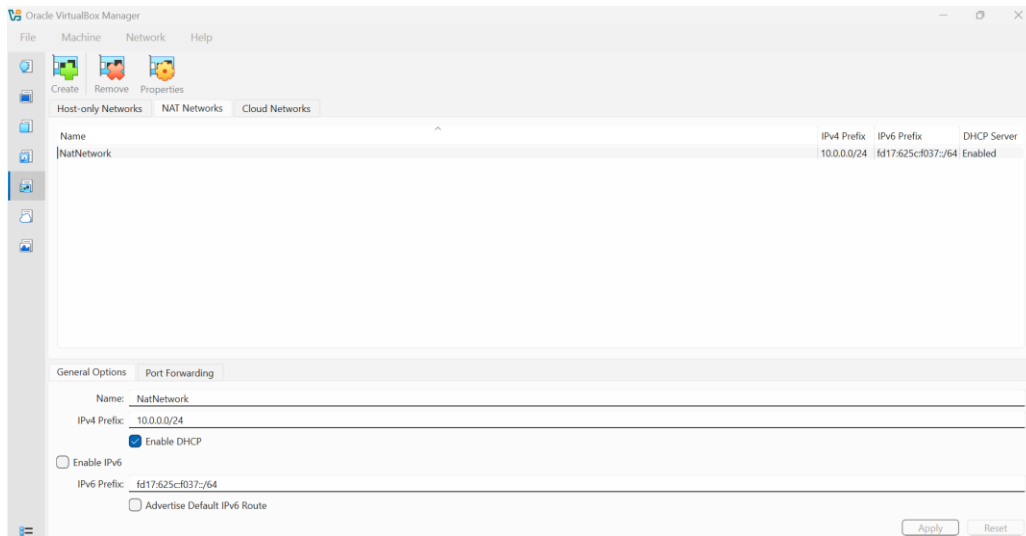
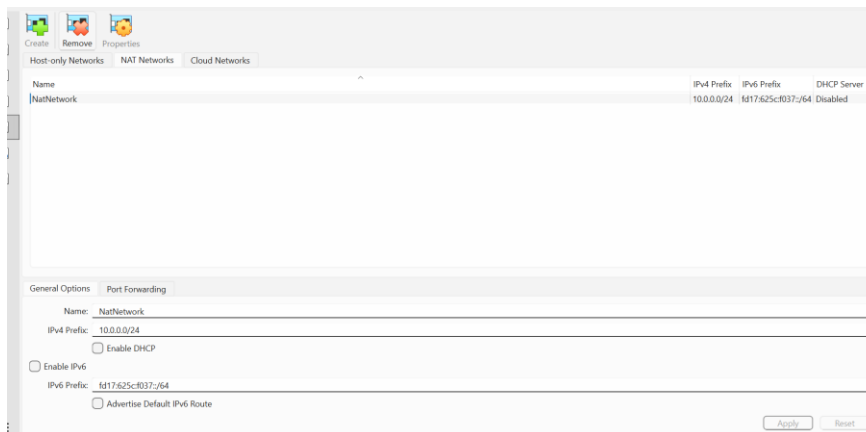


Figure: Kali Linux Network Adapter Configuration

```
(root@kali)-[~]
# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.3/24 brd 10.0.0.255 scope global dynamic noprefixroute eth0
        valid_lft 559sec preferred_lft 559sec
    inet6 fd17:625c:f037:2:b97a:5bfb:4dfa:e0db/64 scope global dynamic nopref
ixroute
        valid_lft 86069sec preferred_lft 14069sec
    inet6 fe80::fc10:7c36:5c40:5dc2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state
DOWN group default
    link/ether f2:96:28:c4:f7:f4 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
```

Figure: Kali Linux Network Interface on 10.0.0.0/24 Subnet



```
(root@kali)-[~]
# ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=36.0 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=23.8 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=25.7 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=24.2 ms

— 8.8.8.8 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 23.770/27.426/36.046/5.024 ms

(root@kali)-[~]
# ip route
default via 10.0.0.1 dev eth0 proto dhcp src 10.0.0.3 metric 100
10.0.0.0/24 dev eth0 proto kernel scope link src 10.0.0.3 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown

(root@kali)-[~]
#
```

```
(root@kali)-[~]
# nmcli connection show
NAME                UUID                                TYPE      DEVICE
Wired connection 1  13145628-be0a-3498-ac18-9c40899fa5cd ethernet  eth0
lo                  2fd10cb9-de5f-4689-a60e-6c26952cb225 loopback   lo
docker0            074b87f7-0a47-47c0-a8fe-9645b0298a89 bridge     docker0
```

```
root@kali: ~  
Session Actions Edit View Help  
(root@kali)-[~]  
# nmcli connection modify "Wired connection 1" ipv4.method auto ipv4.addresses "" ipv4.gateway "" ipv4.dns ""  
(root@kali)-[~]  
# nmcli connection up "Wired connection 1"  
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/8)  
(root@kali)-[~]  
#
```

```
(root@kali)-[~]  
# ip route  
default via 10.0.0.1 dev eth0 proto dhcp src 10.0.0.3 metric 100  
10.0.0.0/24 dev eth0 proto kernel scope link src 10.0.0.3 metric 100  
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown  
(root@kali)-[~]  
# nmcli connection modify "Wired connection 1" ipv4.method manual ipv4.addresses 10.0.0.2/24 ipv4.gateway 10.0.0.1 ipv4.dns 8.8.8.8
```

```
root@kali: ~  
Session Actions Edit View Help  
(root@kali)-[~]  
# nmcli connection show "Wired connection 1" | grep ipv4  
ipv4.method: manual  
ipv4.dns: 8.8.8.8  
ipv4.dns-search: --  
ipv4.dns-options: --  
ipv4.dns-priority: 0  
ipv4.addresses: 10.0.0.2/24  
ipv4.gateway: 10.0.0.1  
ipv4.routes: --  
ipv4.route-metric: -1  
ipv4.route-table: 0 (unspec)  
ipv4.routing-rules: --  
ipv4.replace-local-rule: -1 (default)  
ipv4.dhcp-send-release: -1 (default)  
ipv4.routed-dns: -1 (default)  
ipv4.ignore-auto-routes: no  
ipv4.ignore-auto-dns: no  
ipv4.dhcp-client-id: --  
ipv4.dhcp-iaid: --  
ipv4.dhcp-dscp: --  
ipv4.dhcp-timeout: 0 (default)  
ipv4.dhcp-send-hostname-deprecated: yes  
ipv4.dhcp-send-hostname: -1 (default)  
ipv4.forwarding: -1 (default)  
ipv4.dhcp-hostname: --  
ipv4.dhcp-fqdn: --
```

```
ipv4.dhcp-hostname-flags:      0x0 (none)
ipv4.never-default:            no
ipv4.may-fail:                 yes
ipv4.required-timeout:         -1 (default)
ipv4.dad-timeout:              -1 (default)
ipv4.dhcp-vendor-class-identifier: --
ipv4.dhcp-ipv6-only-preferred: -1 (default)
ipv4.link-local:               0 (default)
ipv4.dhcp-reject-servers:      --
ipv4.auto-route-ext-gw:         -1 (default)
ipv4.shared-dhcp-range:         --
ipv4.shared-dhcp-lease-time:    0 (default)
```

```
(root@kali)-[~]
#
```

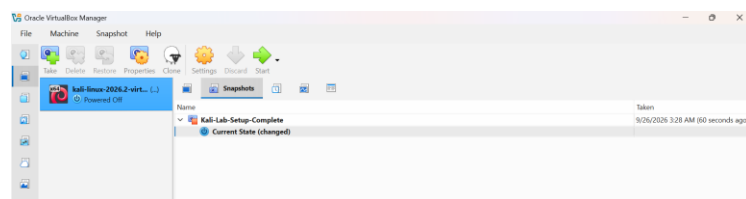
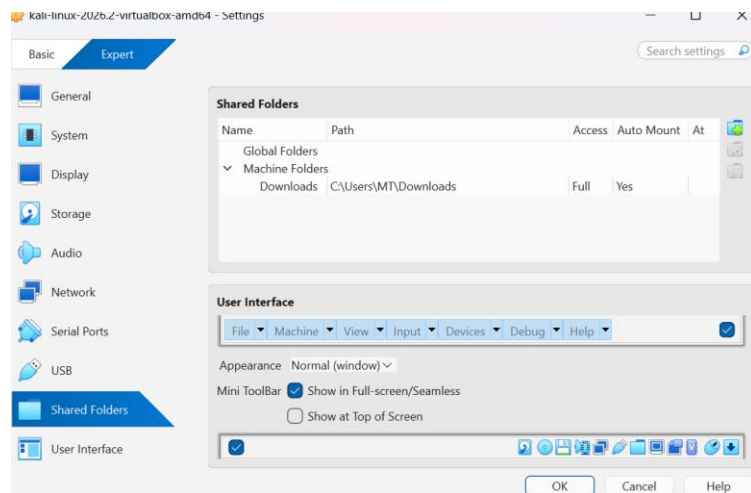
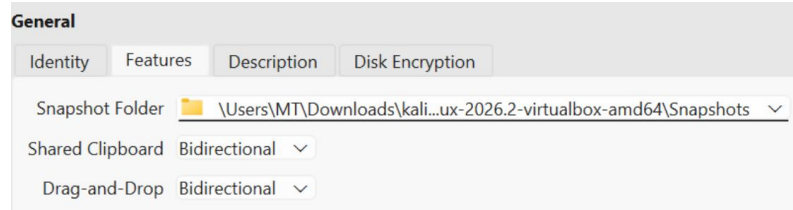
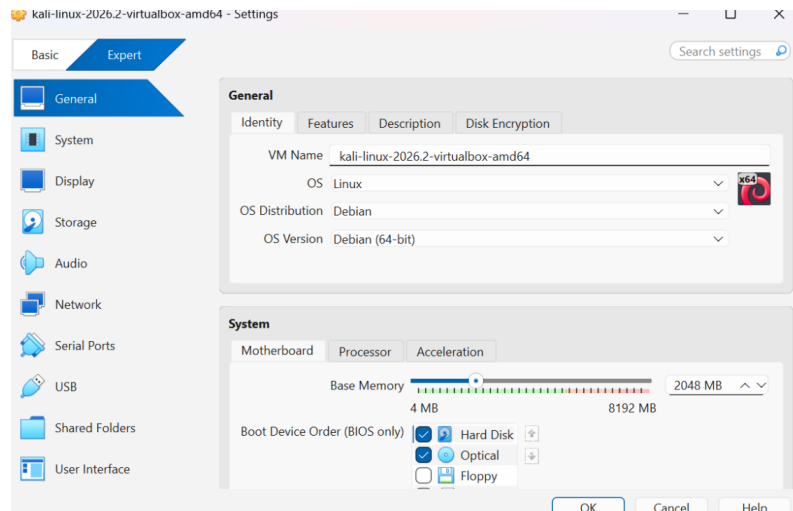
```
(root@kali)-[~]
# nmcli connection modify "Wired connection 1" ipv4.method manual ipv4.addresses 10.0.0.2/24 ipv4.gateway 10.0.0.1 ipv4.dns 8.8.8.8
```

```
(root@kali)-[~]
# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.2/24 brd 10.0.0.255 scope global noprefixroute eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::fc10:7c36:5c40:5dc2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether ea:bc:a6:b3:a2:c8 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
```

```
(root@kali)-[~]
# ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=31.9 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=22.8 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=24.4 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=23.7 ms

— 8.8.8.8 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 22.831/25.707/31.881/3.608 ms
```

```
(root@kali)-[~]
#
```



Conclusion:

The cybersecurity testing lab environment was successfully set up using Oracle VirtualBox and Kali Linux. The required network configuration, Kali Linux IP address 10.0.0.2/24, and Internet connectivity were successfully configured and verified. Shared Clipboard, Drag and Drop, and the downloads shared folder were also configured. Finally, a snapshot was created to preserve the completed lab environment. The setup is now ready for future cybersecurity and ethical hacking practical exercises.